1. Describe one result you found interesting doing your task.

One result I found interesting was that the selected neural network could predict the TGLF fluxes well on both the random test set and the separate SAT3 cases. Its R-squared values ranged from 0.949 to 0.972 on the 750 random test cases and from 0.949 to 0.996 on the 43 SAT3 cases. The SAT3 check was especially useful because those cases came from controlled scans rather than the random sampling used for training. However, the model missed the isotope-order reversal in case c even though it matched the other seven complete triplets. This showed me why checking physical trends matters in addition to reporting a high average score.

2. Describe one thing you struggled with doing your task.

One difficulty in this task was handling the different scales and signs of the outputs without making the comparison misleading. The heat fluxes span a wide range, so predicting their logarithms was a reasonable option, but particle flux can be negative and cannot use the same transformation. I also found four cases with nonpositive ion heat flux and excluded them from the modeling data. To compare the network variants fairly, I evaluated their predictions after converting them back to the original units. Keeping the training, validation, and test sets separate was also important because choosing settings based on test results would make the reported accuracy too optimistic.

3. Identify one choice that you made in your investigation. Why did you make this choice?

One choice I made was to focus on neural networks because my groupmates had already found that they performed best among the model families they compared. I built on that work by comparing network size, depth, and output transformations, then checking whether averaging three networks improved validation error. I kept a training-mean predictor as the simple baseline required by the assignment. Repeating the full comparison with random forests and other models would also have been reasonable, but focusing on neural networks gave my contribution a more specific purpose. I used the validation set to choose the model and reserved the random test set and SAT3 cases for evaluating the final choice.